

A Brain-Computer Interface Control System Design Based on Deep Learning

Third Year Individual Project Final Presentation
Zheng Xu | Student ID: 11314389 | Supervisor: Alex Casson
May 2026

Objective of This Presentation

1

Why MI-BCI robotic control is difficult

2

Which datasets and methods were selected

3

How the EEGTransformer decoder works

4

How channel reduction supports OpenBCI

5

How DQN closes the control loop

6

What the results, limits, and next steps show

Main message: the contribution is an integrated decoder -> controller -> deployment-oriented pipeline.

Project Aim and Problem

Project aim

Translate motor-imagery EEG into robotic arm commands using deep learning and reinforcement learning.

1

Non-stationary EEG

Different users and sessions shift the signal distribution.

2

Open-loop fragility

A single misclassified epoch can push the arm in the wrong direction.

3

Deployment gap

High-density 64-channel studies do not map cleanly to low-cost hardware.

The project therefore evaluates an offline/simulated perception -> decision -> action loop, not classification accuracy alone.

Background: Motor-Imagery BCI

1

EEG

Scalp electrodes record microvolt brain activity

2

Motor imagery

Labels depend on dataset;
imagined movement becomes
command evidence

3

Mu/Beta rhythms

8-30 Hz activity over sensorimotor
cortex changes

4

Robot command

Predicted intention becomes a
directional control cue

Why motor imagery?

It requires no external stimulus and supports voluntary assistive control, but single-trial EEG is noisy and user-dependent.

Control challenge

A robot must recover from wrong cues over time, so the controller must be closed-loop rather than a fixed label-to-action map.

Literature Review: Method Gap

CSP / FBCSP

Interpretable spatial filters

Sensitive to covariance drift

EEGNet / CNNs

Learn compact spatial-temporal filters

Mostly independent single-trial decisions

EEG-Conformer / ATCNet

Stronger CNN-attention decoders

Classifier output still treated as endpoint

RL for EEG

Useful for feature selection or adaptation

Less often used for actuator control

73.80%

Ours on BCI IV-2a

77.7%

EEG-Conformer

80.5%

ATCNet

Positioning: indicative literature context, not a direct leaderboard; the main contribution is the closed-loop pipeline.

Dataset and Method Choice

BCI IV-2a

22 ch | 4 classes | 9 subjects

Standard multi-class benchmark for literature comparison.

BCI IV-2b

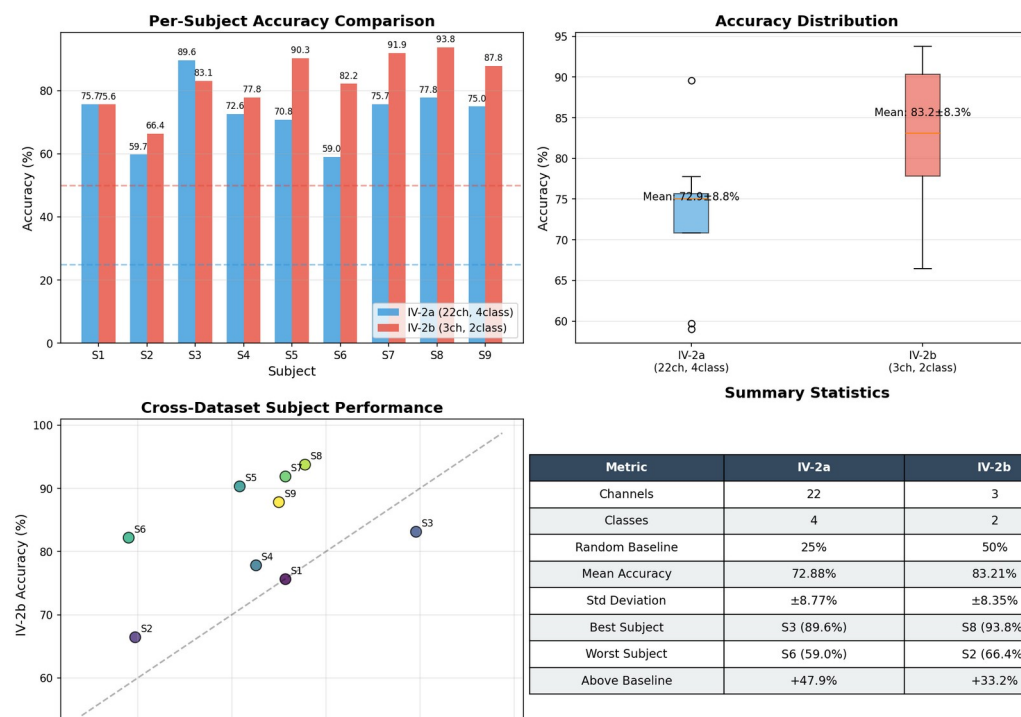
3 ch | 2 classes | 9 subjects

Minimal-channel stress test under strong hardware constraint.

PhysioNet EEGMMIDB

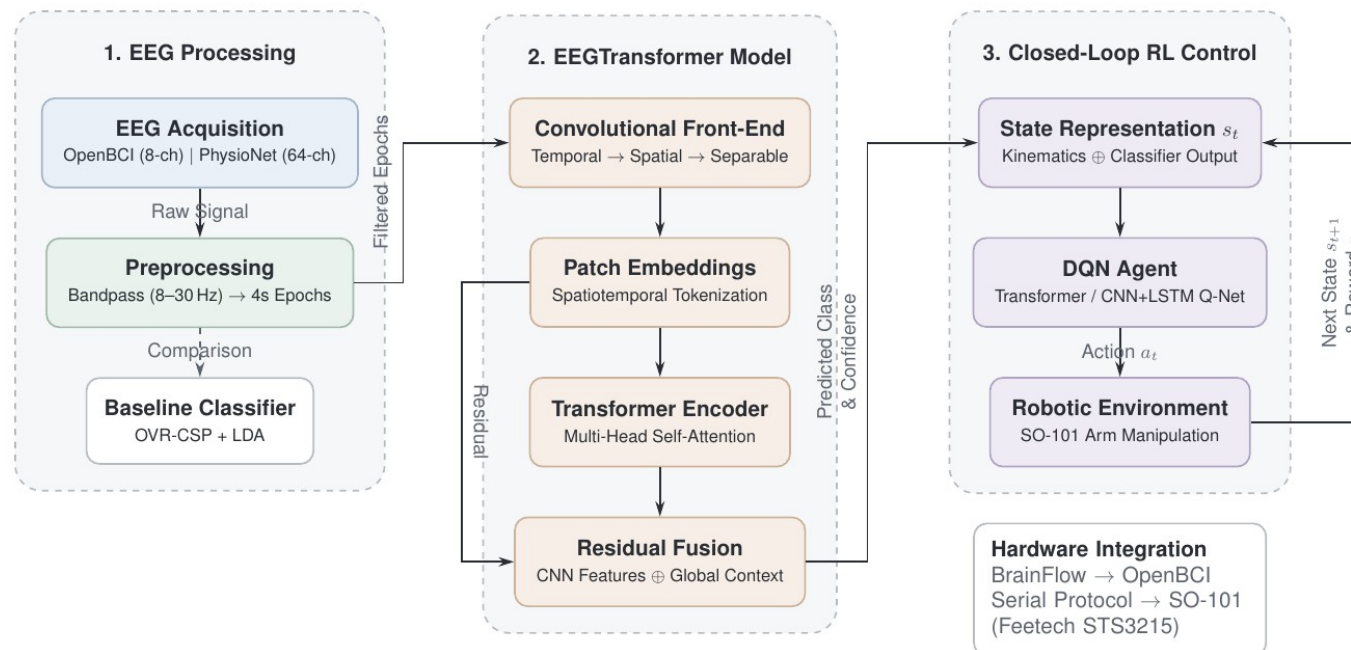
64 ch | 4 classes | 109 subjects

Large subject pool for cross-subject pre-training and fine-tuning.



The datasets are evaluated independently because their montages, class definitions, and published protocols differ.

Overall System Architecture



4-class MI → 4 commands: left, right, up, down

Classifier output is noisy DQN evidence; OpenBCI was interface/synthetic-board validated, not live EEG controlled.

Preprocessing and Evidence

Raw EEG

EDF / MAT loading, channel standardisation

8-30 Hz bandpass

Keeps motor-imagery Mu/Beta information

Epoch + resample

Fixed input tensor for EEGTransformer

Normalise

Training statistics reused on held-out data

+18.44%

5-subject PhysioNet filter ablation

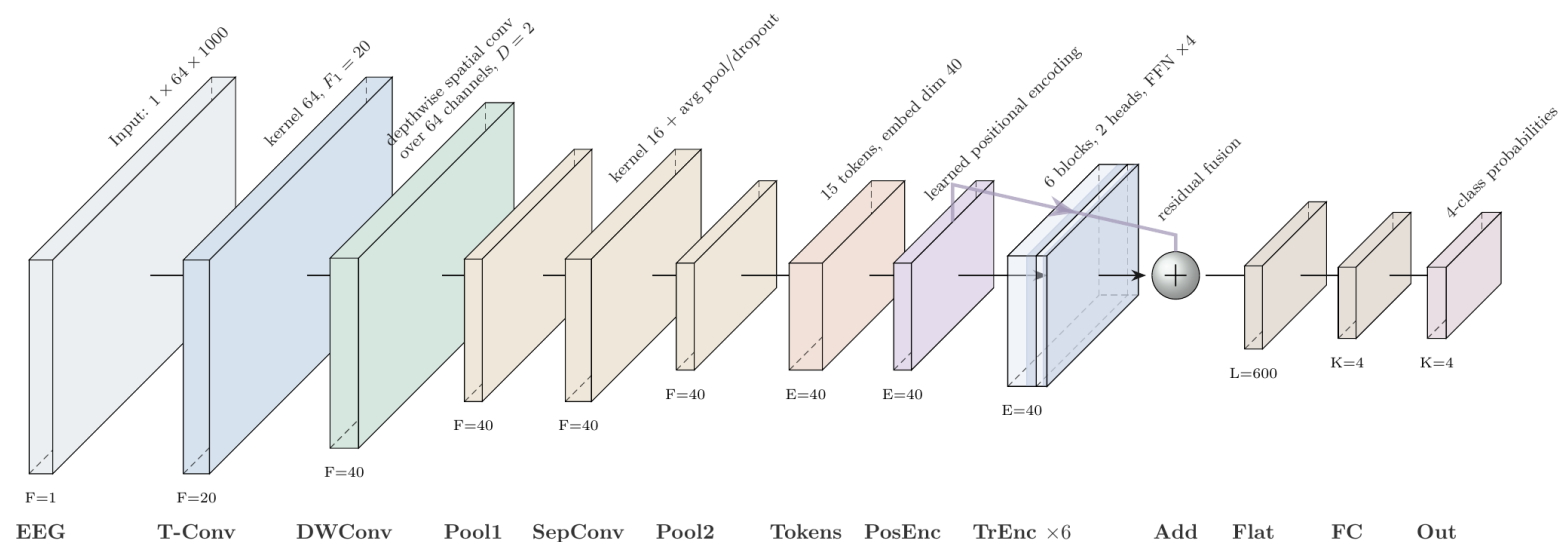
+1.51%

3-subject BCI IV-2a ICA test

Resulting choice

Use a bandpass-only pipeline for the main experiments. ICA is reported as limited-scope evidence, not a universal conclusion.

EEGTransformer Design



CNN front-end

Learns temporal filters and data-driven spatial filters.

Transformer encoder

Models long-range structure across the MI window.

Residual fusion

Combines local EEG features with global context before classification.

64 x 1000

Main PhysioNet input

15 tokens

After pooling

4 classes

MI command output

Training and Transfer Learning

56.54%

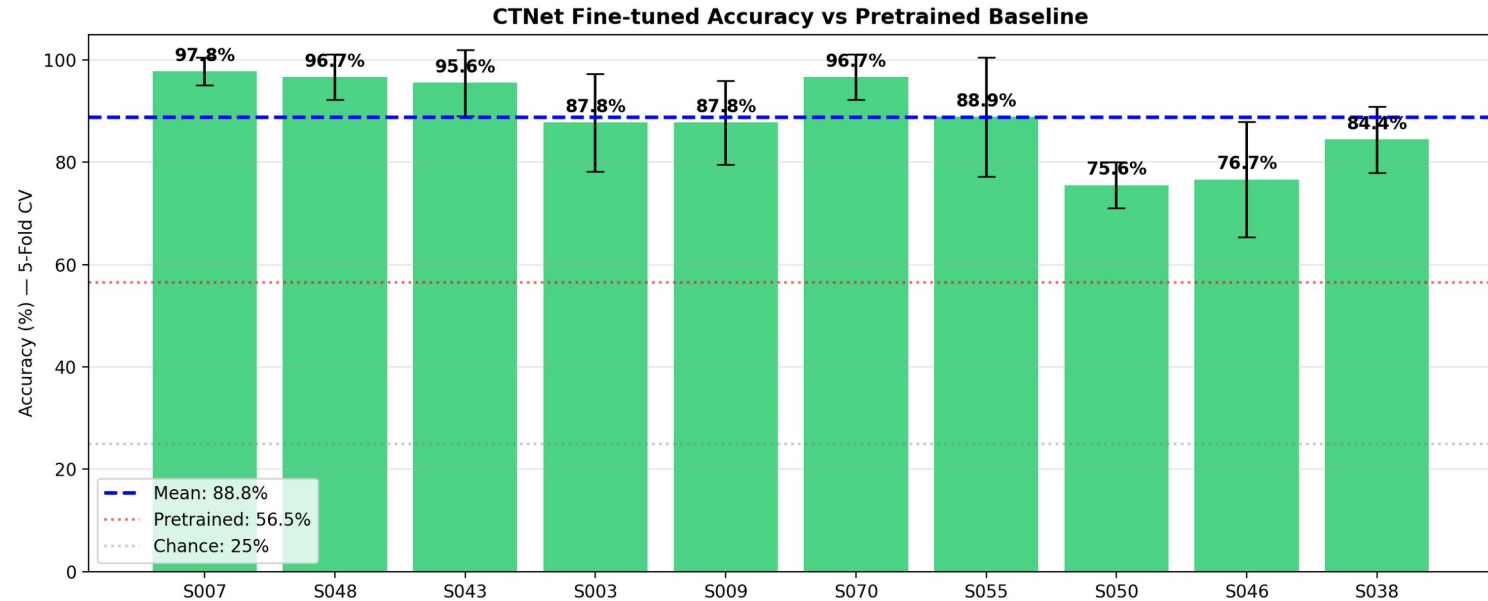
Pooled cross-subject PhysioNet

88.78%

Fine-tuned mean | 10 subjects

+32 pts

Gap reduced in fine-tuning eval



Interpretation: fine-tuning adapts a transferable MI representation; broader full-cohort validation remains future work.

Classification Results: Useful Decoder, Not SOTA

73.80%

BCI IV-2a | 22 ch | 4 class

82.87%

BCI IV-2b | 3 ch | 2 class

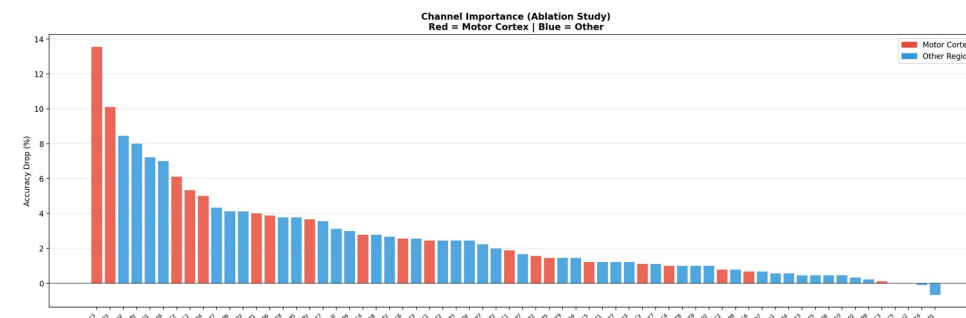
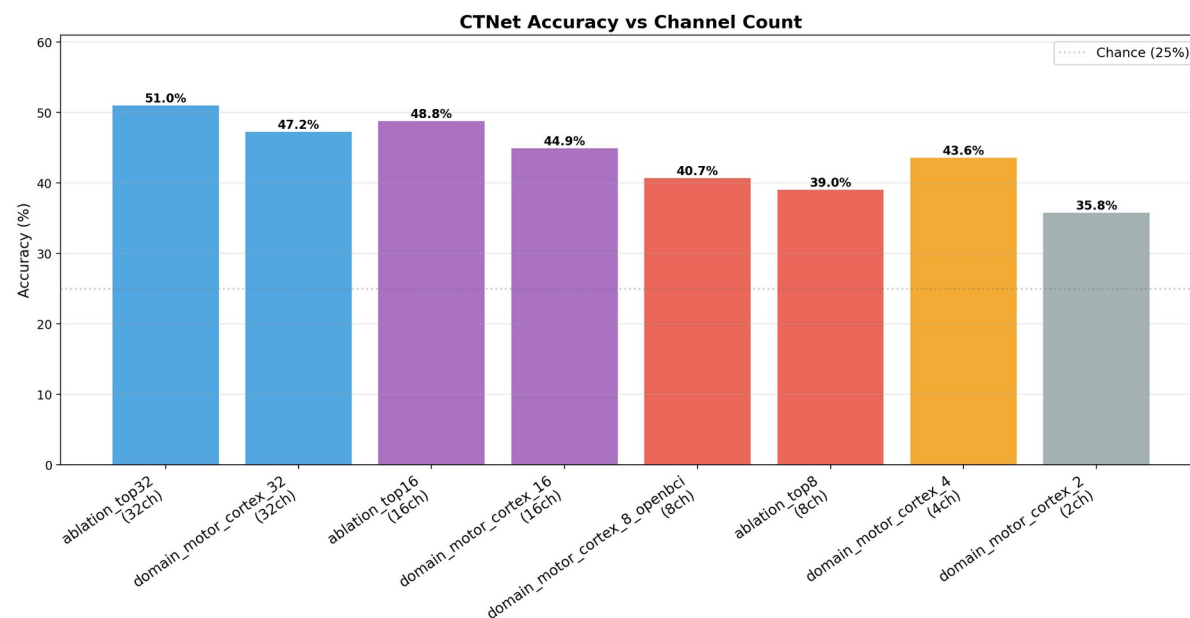
88.78%

PhysioNet | fine-tuned

Literature context: competitive with CNN baselines, below stronger recent attention models; contribution is system integration.

FBCSP + SVM	67.6%	Classical baseline
EEGNet / ShallowConvNet	68-72%	Compact CNN family
EEG-Conformer	77.7%	CNN + Transformer
ATCNet	80.5%	Attention temporal CNN
EEGTransformer (ours)	73.80%	Competitive, system-oriented

OpenBCI-Oriented Channel Reduction



C3

Largest ablation
drop

8 ch

OpenBCI-
compatible montage

72.54%

Fine-tuned accuracy

Boundary: this is offline evidence for an 8-channel montage; native live OpenBCI validation is future work.

Robotic Control and RL Method



State

End-effector position, target position, distance, optional EEG class + confidence

Action

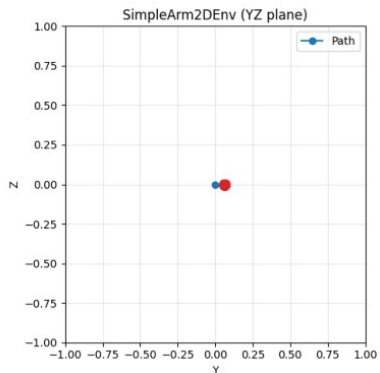
Four discrete directions: left, right, up, down

Reward

Target reaching, distance improvement, step penalty, boundary penalty

Environment

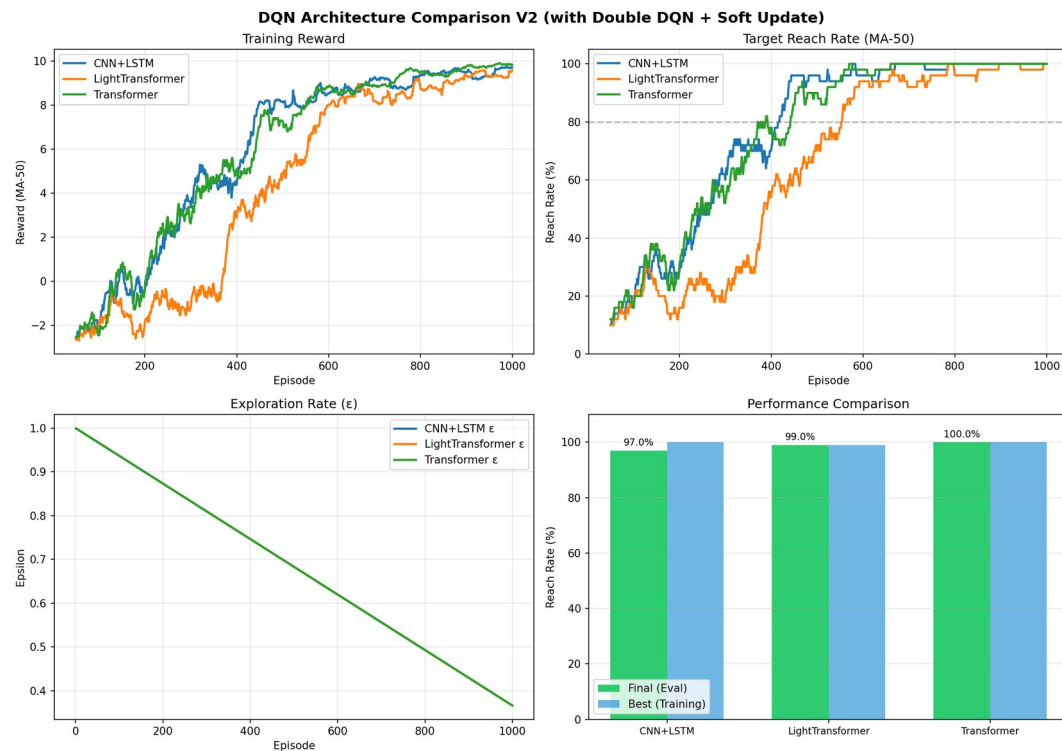
PyBullet / planar reaching simplification before richer SO-101 manipulation



Top: physical SO-101 interface frame.
Bottom: animated control GIF plays in
PPTX; PDF shows a static frame.

Controller role: use the current state to correct transient EEG misclassifications instead of blindly executing each label.

RL Results



97%

CNN+LSTM final reach rate

99%

Light Transformer reach rate

100%

2D benchmark reach rate

10.39

Transformer final reward

Under this simulated reaching protocol, the full Transformer scores highest; Light Transformer is a practical trade-off.

End-to-End Results

82.22%

EEG accuracy on 630 trials

98.7%

EEG-aware reach rate

99.0%

State-only reach rate

668s

With EEG state

1130s

Without EEG state

Why control can exceed decoding

- The DQN does not execute each class label blindly.
- It observes state, target distance, class, and confidence over multiple steps.
- Reward feedback allows wrong directional cues to be corrected.

What EEG added in this test

- Final reach rate was similar: 98.7% with EEG vs 99.0% state-only.
- Training was faster with EEG evidence: 668 s vs 1130 s.
- This supports usefulness, not proof that EEG is essential in the simplified task.

Main interpretation: closed-loop DQN compensates for imperfect EEG; EEG evidence mainly accelerates learning in this simulation.

Discussion, Conclusion, and Next Steps

What was shown

- EEGTransformer works competitively: 73.80%, 82.87%, 88.78%
- Fine-tuning reduces cross-subject weakness: 56.54% -> 88.78%
- 8-channel montage remains usable offline: 72.54%
- End-to-end simulation: 98.7% EEG-aware, 99.0% state-only

What was not shown

- No live human-subject EEG control
- BrainFlow was synthetic-board interface validation
- 8-channel OpenBCI result is offline, not native live deployment
- 4 s online window may be too slow for real-time use

Next steps

- Ethics-approved live-user OpenBCI study
- Native real-time 8-channel inference path
- Shorter overlapping windows or asynchronous MI
- Richer SO-101 manipulation beyond planar reaching

Validation boundary: EEG classification is offline; control is simulated; live human closed-loop testing requires ethical approval.